1. Aim of the practical: Creating and managing users and groups using user add, userdel, groupadd and passwd. Controlling Access to Files with Access Control Lists (ACL) using chmod, chgrp and chown

2. Objective: The objective of this experiment is to learn how to manage users, groups, and control file access using Linux commands like useradd, groupadd, chmod, chgrp, and chown.

3. Hardware and Software Requirements:

a) **Disk Drive**: Physical disk (HDD/SSD) with enough space.
Computer System: Host system running virtualization software.
b) **Operating System**: OS on the virtual machine (e.g., Fedora).

4. Commands/Steps:

Step-1: Creating a new user using sudo useradd username.

```
nikhilyadav@fedora:~$ sudo useradd softwaredeveloper
[sudo] password for nikhilyadav:
```

Step-2: Setting password for the new user

```
nikhilyadav@fedora:~$ sudo passwd softwaredeveloper
New password:
Retype new password:
passwd: password updated successfully
```

Step-3: Now Creating a new group using sudo groupadd Developers

```
nikhilyadav@fedora:~$ sudo groupadd bros
nikhilyadav@fedora:~$ sudo usermod -aG bros softwaredeveloper
nikhilyadav@fedora:~$ groups bros
groups: ‘bros’: no such user
```

Step-4: Now adding badmas user into coolGroup

```
nikhilyadav@fedora:~$ sudo groupadd bros
nikhilyadav@fedora:~$ sudo usermod -aG bros softwaredeveloper
nikhilyadav@fedora:~$ groups bros
groups: ‘bros’: no such user
```

```
nikhilyadav@fedora:~$ getent group bros
bros:x:1002:softwaredeveloper
```

Step-5: Now, Checking the developer groups.

```
nikhilyadav@fedora:~$ groups softwaredeveloper
softwaredeveloper : softwaredeveloper bros
```

Step-6: Now creating a new file named demo.txt

```
nikhilyadav@fedora:~$ touch demo.txt
nikhilyadav@fedora:~$ ls
demo.txt  Documents  monday.txt  Pictures  Templates
Desktop   Downloads  Music       Public    Videos
```

Step-7: Checking the permission and owner of the file.

```
nikhilyadav@fedora:~$ ls -l demo.txt
-rw-r--r--. 1 nikhilyadav nikhilyadav 0 Sep 26 23:20 demo.txt
```

Step-8: Changed the owner of the file.

```
nikhilyadav@fedora:~$ sudo chown softwaredeveloper:bros demo.txt
nikhilyadav@fedora:~$ ls -l demo.txt
-rw-r--r--. 1 softwaredeveloper bros 0 Sep 26 23:20 demo.txt
```

Step-9: The command sudo chmod 660 myfile.txt sets the permissions of myfile.txt to allow read and write access for the owner and group, but no access for others.

```
nikhilyadav@fedora:~$ sudo chmod 660 demo.txt
nikhilyadav@fedora:~$ ls - demo.txt
ls: cannot access '-': No such file or directory
demo.txt
nikhilyadav@fedora:~$ ls -l demo.txt
-rw-rw----. 1 softwaredeveloper bros 0 Sep 26 23:20 demo.txt
```

Step-10: chomd is used to add read and write permissions for the user on demo1.txt.

```
nikhilyadav@fedora:~$ sudo chmod u+rwx demo.txt
nikhilyadav@fedora:~$ ls -l demo.txt
-rwxrw----. 1 softwaredeveloper bros 0 Sep 26 23:20 demo.txt
```

Step-11: command is used to give read permission to a specific user on demo1.txt.

```
nikhilyadav@fedora:~$ sudo setfacl -m u:softwaredeveloper:rw demo.txt
nikhilyadav@fedora:~$ ls -l demo.txt
-rwxrw----+ 1 softwaredeveloper bros 0 Sep 26 23:20 demo.txt
```

Step-12: Second command is used to display the ACL entries for file.txt.

```
nikhilyadav@fedora:~$ getfacl demo.txt
# file: demo.txt
# owner: softwaredeveloper
# group: bros
user::rwx
user:softwaredeveloper:rw-
group::rw-
mask::rw-
other::---
```

Step-13: Deleting the user, Deleting the group now

```
nikhilyadav@fedora:~$ sudo userdel softwaredeveloper
[sudo] password for nikhilyadav:
Sorry, try again.
[sudo] password for nikhilyadav:
nikhilyadav@fedora:~$ sudo groupdel bros
```

5. Result:

```
nikhilyadav@fedora:~$ getfacl demo.txt
# file: demo.txt
# owner: 1001
# group: 1002
user::rwx
user:1001:rw-
group::rw-
mask::rw-
other::---
```

6. Learning outcomes (What I have learned):

a) Understand how to create, modify, and delete user accounts using useradd and userdel.

b) Learn how to organize users into groups using groupadd and manage group memberships.

c) Master the use of chmod to set and modify file permissions.

Teacher Signature